

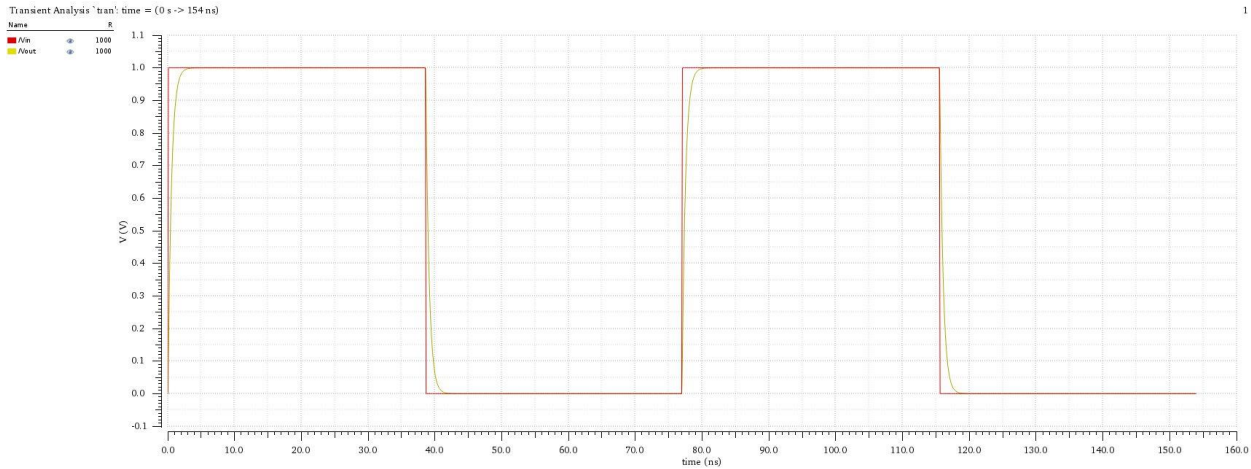
Lab 01 Report

LPF Simulation and MOSFET Characteristics

PART 1: Low Pass Filter Simulation (LPF) [period=77ns, time constant=0.5ns]

1. Transient Analysis

Transient Analysis Results (Two Periods)



Rise and Fall Time (10% to 90%) Comparison

The charging equation for the RC low-pass filter responding to a step input V_{in} is:

$$V_{out}(t) = V_{in}(1 - e^{-t/\tau})$$

To find the time t_{10} when the output reaches 10% (0.1) of V_{in} :

$$0.1V_{in} = V_{in}(1 - e^{-t_{10}/\tau}) \implies t_{10} = -\tau \ln(0.9) \approx 0.105\tau$$

To find the time t_{90} when the output reaches 90% (0.9) of V_{in} :

$$0.9V_{in} = V_{in}(1 - e^{-t_{90}/\tau}) \implies t_{90} = -\tau \ln(0.1) \approx 2.303\tau$$

The rise time t_r is the difference between these two points:

$$t_r = t_{90} - t_{10} = \tau(\ln(0.9) - \ln(0.1)) = \tau \ln(9) \approx 2.197\tau \approx 2.2\tau$$

Substituting the given time constant $\tau = 0.5$ ns:

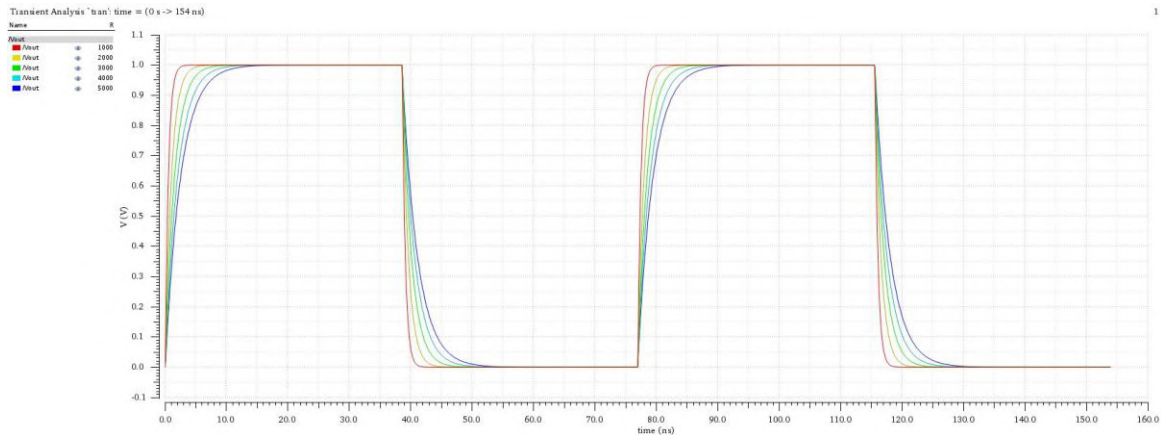
$$t_r \approx 2.2 \times 0.5 \text{ ns} = 1.1 \text{ ns}$$

The fall time t_f (from 90% down to 10%) follows the exact same derivation using the discharging equation $V_{out}(t) = V_{in}e^{-t/\tau}$, yielding the identical relationship:

$$t_f \approx 2.2\tau = 1.1 \text{ ns}$$

Parameter	Simulation Result	Analytical Result
Rise Time	1.104 ns	1.1 ns
Fall Time	1.102 ns	1.1 ns

Parametric Sweep (R=1:1:5kΩ) Overlaid Results

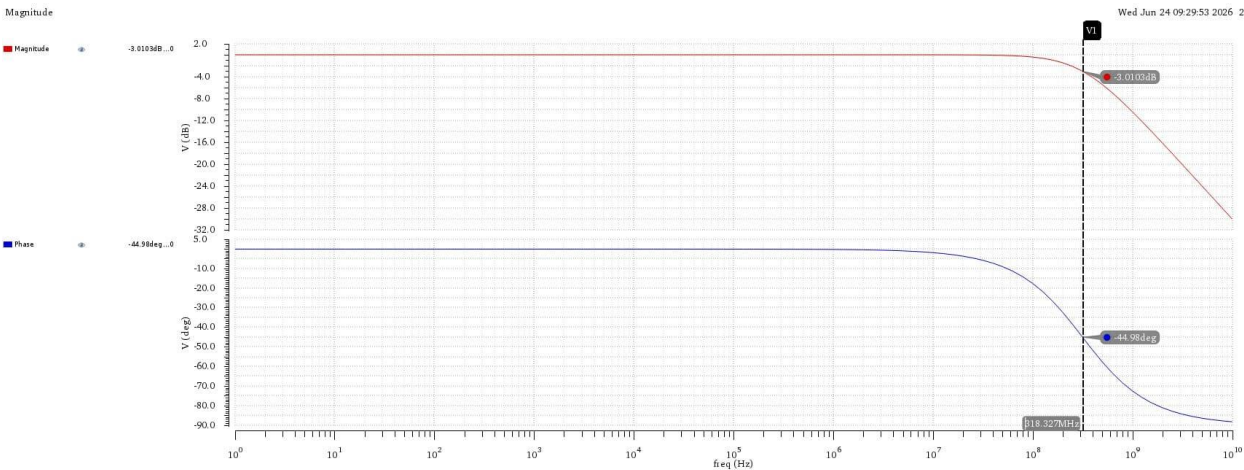


Comments on Parametric Sweep Results

Because the rise and fall times are directly proportional to the time constant ($tr \approx 2.2\tau$), increasing the resistance drastically slows down the charging and discharging rate of the capacitor. This is visually evident in the plot: the 1 kΩ curve (red) is the steepest and fastest, while the 5 kΩ curve (dark blue) is the most sluggish.

2. AC Analysis

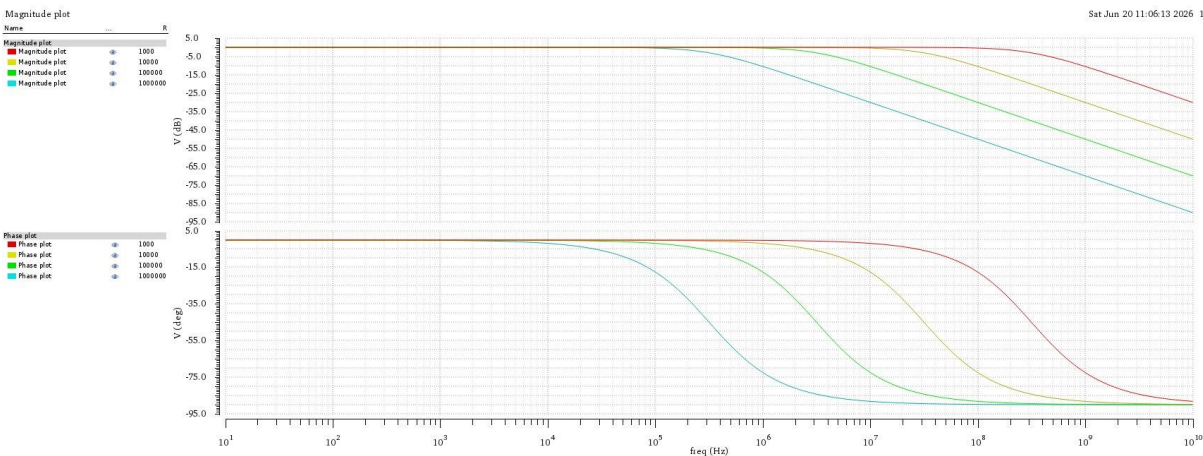
Bode Plot (Magnitude and Phase)



DC Gain and 3dB Bandwidth Comparison

Parameter	Simulation Result	Analytical Result
DC Gain	0 dB	0 dB
3dB Bandwidth	317.6 MHz	318.3 MHz

Parametric Sweep (R=1, 10, 100, 1000kΩ) Overlaid Results



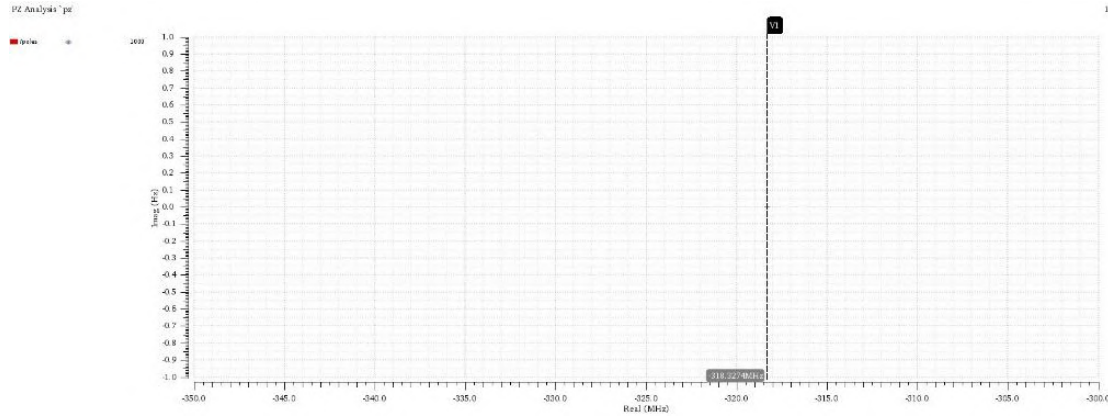
Comments on Parametric Sweep Results

Inverse Relationship with Bandwidth: The plots perfectly demonstrate that the 3dB bandwidth (corner frequency) is inversely proportional to the resistance ($f_c = 1 / 2\pi RC$). As the resistance is multiplied by 10 in each sweep step, the "knee" of the magnitude plot shifts to the left by exactly one decade on the frequency axis.

Point	Test	Output	Nominal
Parameters: R=1k			
1	LPF_Mod:LPF_Mod:1	Bandwidth	317.6M
Parameters: R=10k			
2	LPF_Mod:LPF_Mod:1	Bandwidth	31.76M
Parameters: R=100k			
3	LPF_Mod:LPF_Mod:1	Bandwidth	3.176M
Parameters: R=1M			
4	LPF_Mod:LPF_Mod:1	Bandwidth	317.6k

3. Pole Zero Analysis

Pole Zero Analysis Results



Pole Frequency vs. AC Analysis Bandwidth Comparison

The transfer function $H(s)$ of the RC low-pass filter:

$$H(s) = \frac{V_{out}(s)}{V_{in}(s)} = \frac{\frac{1}{sC}}{R + \frac{1}{sC}} = \frac{1}{1 + sRC}$$

To extract the pole, set the denominator of the transfer function to zero:

$$1 + sRC = 0 \implies s_p = -\frac{1}{RC}$$

Substituting the time constant $\tau = RC = 0.5$ ns:

$$s_p = -\frac{1}{0.5 \times 10^{-9}} = -2 \times 10^9 \text{ rad/s}$$

Converting the pole magnitude from radians per second to frequency (f_c) in Hertz:

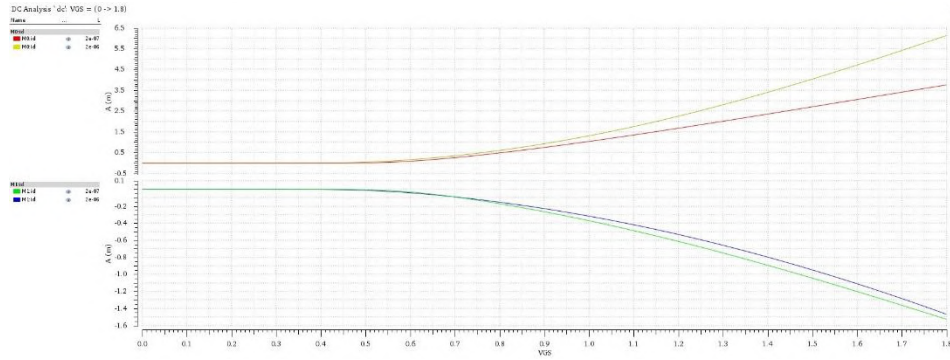
$$f_c = \frac{|s_p|}{2\pi} = \frac{1}{2\pi RC}$$
$$f_c = \frac{2 \times 10^9}{2\pi} \approx 318.3 \text{ MHz}$$

The pole is exactly the analytical 3dB bandwidth.

PART 2: MOSFET Characteristics [W/L=32]

1. ID vs VGS

ID vs VGS Characteristics (Overlaid for Short & Long Channel)



In Long Channel, Current is quadratic as electric field is relatively weak, so carrier velocity increases linearly with electric field. However in Short Channel, It is linear.

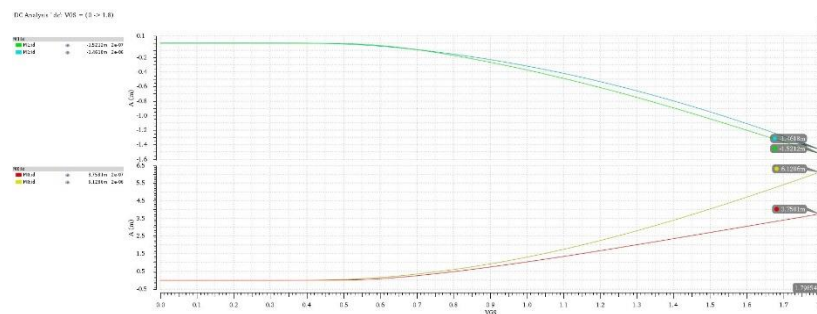
Long channel has higher current as sub-micron processes suffer from velocity saturation where at $E_{cr} = \frac{V_{DS}}{L}$, The velocity of carriers (either holes in PMOS or electrons in NMOS) saturates before reaching pinch off, leading to less current.

NMOS is more affected by short channel effects compared to PMOS, as more electrons pass from source to drain compared to the holes.

NMOS has higher current than PMOS as mobility of electrons is higher than that of holes.

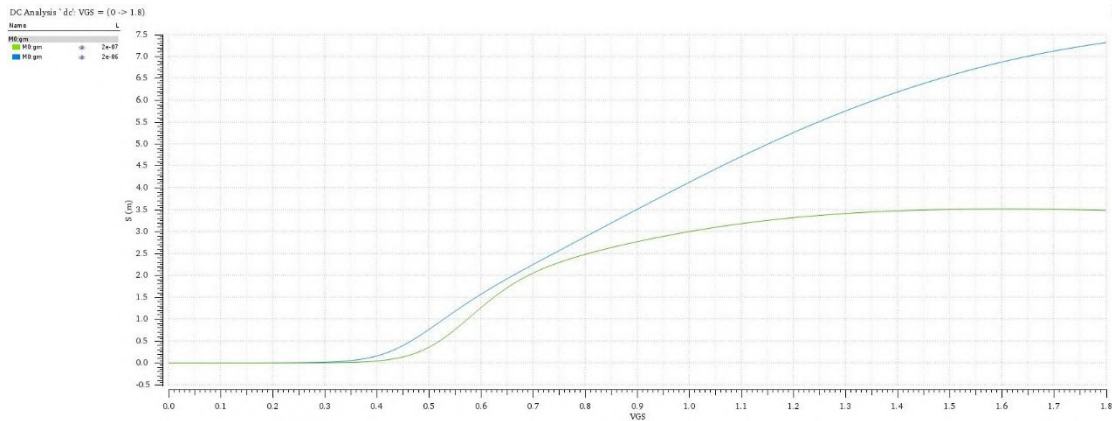
@ $V_{GS}=1.8V$, $\left| \frac{I_{Dn}}{I_{Dp}} \right| = 2.47$ in short channel, and $\left| \frac{I_{Dn}}{I_{Dp}} \right| = 4.19$ in long channel.

NMOS is severely affected by short channel effects as their carriers(electrons)reach velocity saturation at lower E . Therefore, It deviates from the square law earlier.



2. gm vs VGS

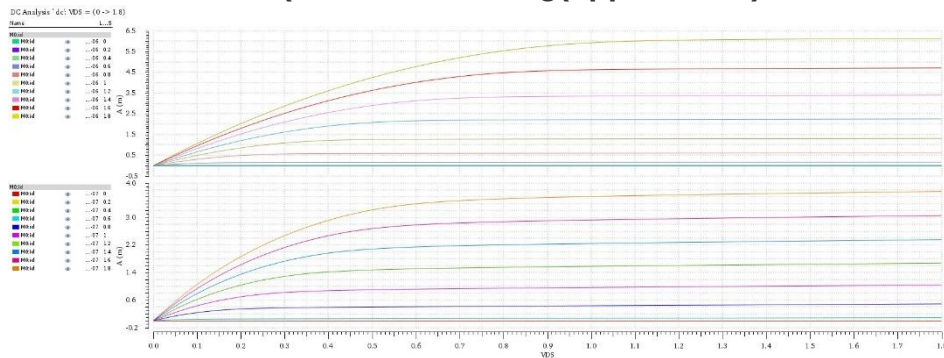
gm vs VGS Characteristics (Overlaid for Short & Long Channel)



gm increases linearly in long channel devices, as current increases quadratically (follows square law) after pinch off. On the other hand, gm saturates in short channel devices, as current increases linearly. ($g_m = \frac{\partial I_D}{\partial V_{GS}}$)

3. ID vs VDS

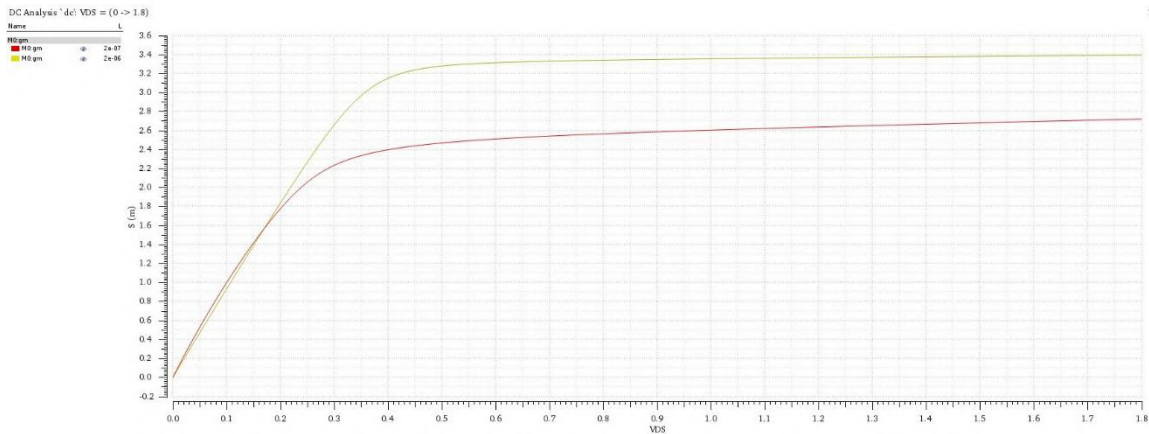
ID vs VDS Characteristics (Overlaid for Long(upper curve) & Short Channel(lower))



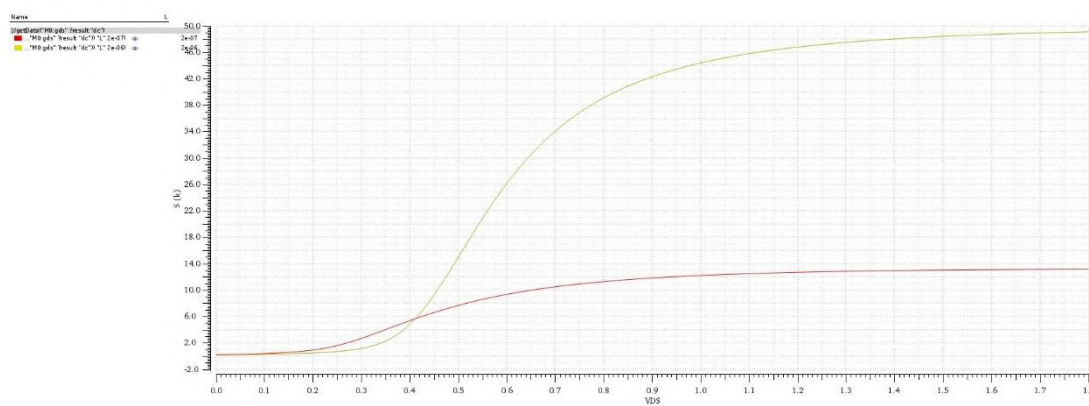
Long channel has higher current. The short-channel device has a significantly higher slope in the saturation region. This is caused by Channel Length Modulation (CLM). As V_{DS} increases, the drain depletion region expands and reduces the effective length of the channel (L_{eff})

4. gm and ro in Triode and Saturation

gm vs VDS Characteristics



ro vs VDS Characteristics



Linearity in Triode: In the triode region, g_m goes up in a straight line as V_{DS} increases. This happens because the current relies heavily on the drain voltage in this region

$$g_m \approx \mu C_{ox} \left(\frac{W}{L} \right) V_{DS}$$

gm Saturation: Once the transistor hits pinch-off and enters the saturation region, the current mostly stops reacting to V_{DS} . Because of this, g_m stops rising, flattens out, and becomes a steady value controlled mostly by V_{GS} .

Preferred Operating Region: Analog amplifiers must operate in the saturation region. This gives you the highest possible g_m . A stable g_m is required to keep the amplifier's gain steady.

Onset vs g_m : Unlike g_m , r_o doesn't flatten immediately at pinch-off; it is still rising at the saturation boundary.

Edge Operation: Avoid the exact edge of saturation. A tiny voltage drop pushes the device into triode, crashing r_o and amplifier gain.

Preferred Region: Bias deep in saturation. This ensures a high, stable r_o for good gain ($A_v = g_m * r_o$).